



Honeywell

APM - UNISIM INTEGRATION AND VALUE PROPOSITION

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AGENDA

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 2. License
3. Unisim Build Installation and case creation
4. Security and Permission for Unisim Case
5. DCOM setting on calculation server
6. UniSim Design data source Configuration
7. Input/Output Variables configuration
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9. Snapshot file – Success or fail
10. Energy Opportunity Dashboard

APM – Unisim Design Overview

APM – UNISIM DESIGN OVERVIEW

- . The key performance parameters of assets that are modeled using USD can be calculated.
- The results of calculation from Unisim Design sent back to Asset Performance Management.
- UniSim Design can be integrated with Asset Performance Management to perform KPI calculations by running a simulator in calculation mode and to perform optimization for reconciliation or process/energy optimization by running a simulator in optimization mode.
- Two type of optimization engines are supported:
 - NOVA
 - Legacy SQP

PRE-REQUISITES

1. Software Compatibility

- UAS 540.1/.2/.3 version is compatible with Unisim R492, R500 and R510.

Note: Although R500 and R510 isn't tested by DE but project team has tested and it's working fine.

UAS Version	Unisim Version
R540.1	R492, R500, R510
R540.2	R492, R500, R510
R532.5	R492

2. License

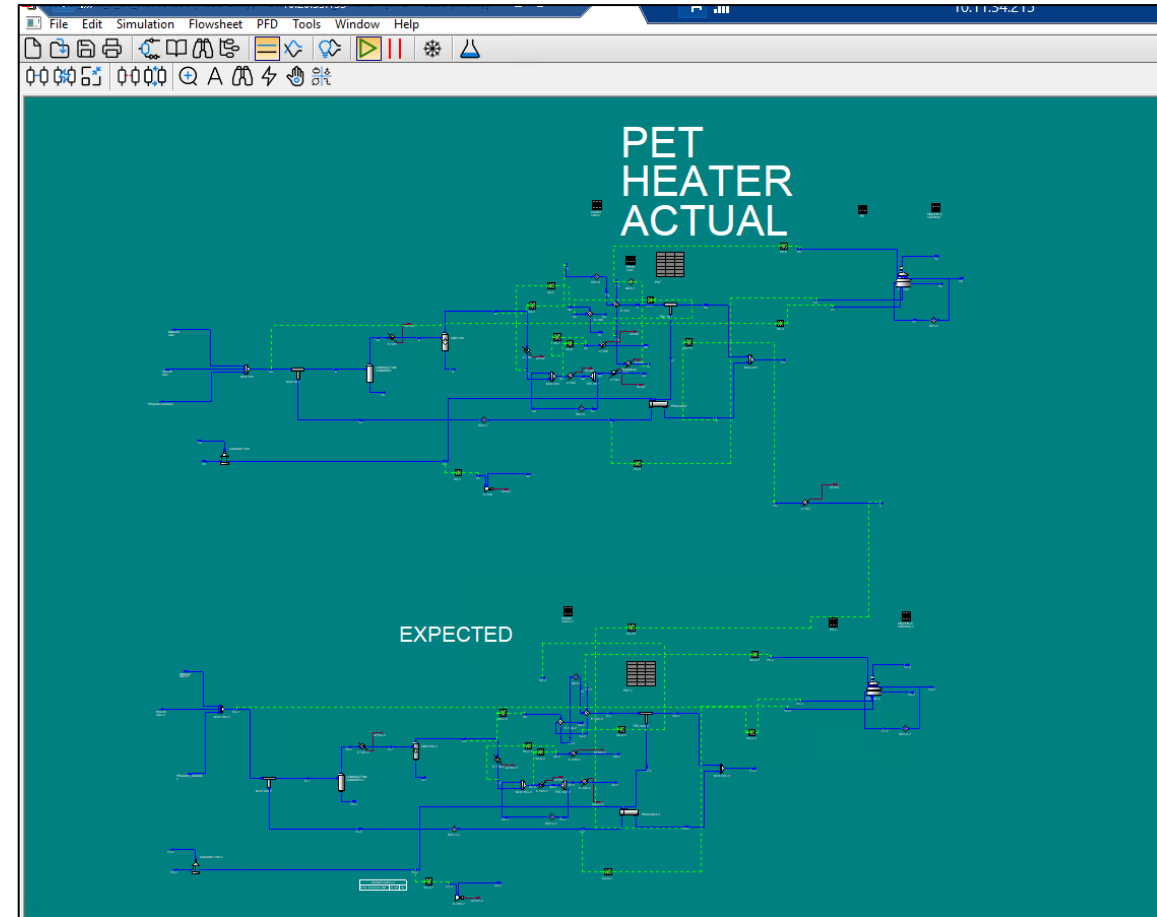
Ensure that APM unisim interface should have below license:

- Sentinel.UniSim Link (SEN) – For APM to Unisim interface.
- UniSimDesign.Rating (USD) – Required for Snapshot file creation for unisim case and maintenance of unisim case for any update.

Unisim Build Installation and case creation

UNISIM BUILD INSTALLATION AND CASE CREATION

1. Ensure that a licensed version of USD simulator (version R492/500/510) is installed on the Calculation server.
2. Unisim model in .usc format created which get converged without any errors with static data.
3. Ensure that tag map all the input variables configured in USD as "Inputs" and output variables configured in USD as "Outputs" to the Asset Performance Management attributes.
4. Ensure that the process data table names are configured for all input attributes written from Asset Performance Management as "Input" and output attributes read by Asset Performance Management as "Output" in the USD model.

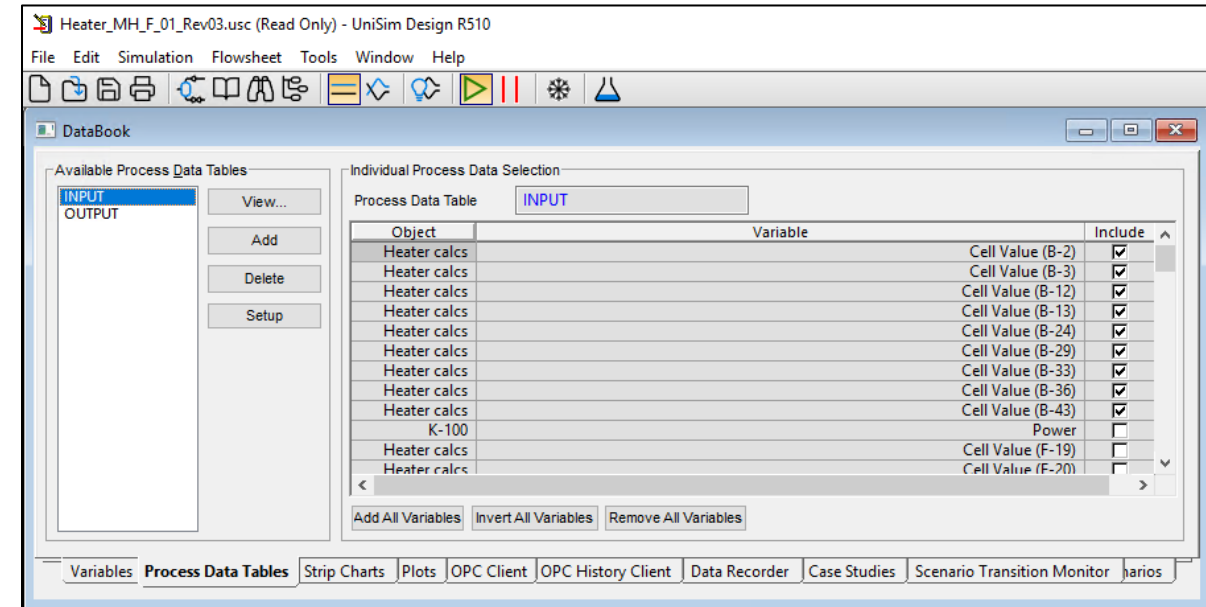


UNISIM BUILD INSTALLATION AND CASE CREATION

1. Check the Input and Output PDT by “Ctrl+D” and select Input → View
2. Ensure that all Attributes should be mapped correctly with Access Mode as “Read/Write”. There should not be any constant/fix value with No tag mapping and “No Transfer” as Access Mode.

NOTE All the variables that you select must be tagged in Asset Performance Management.

Hence, ensure that you select only the variables that you require.



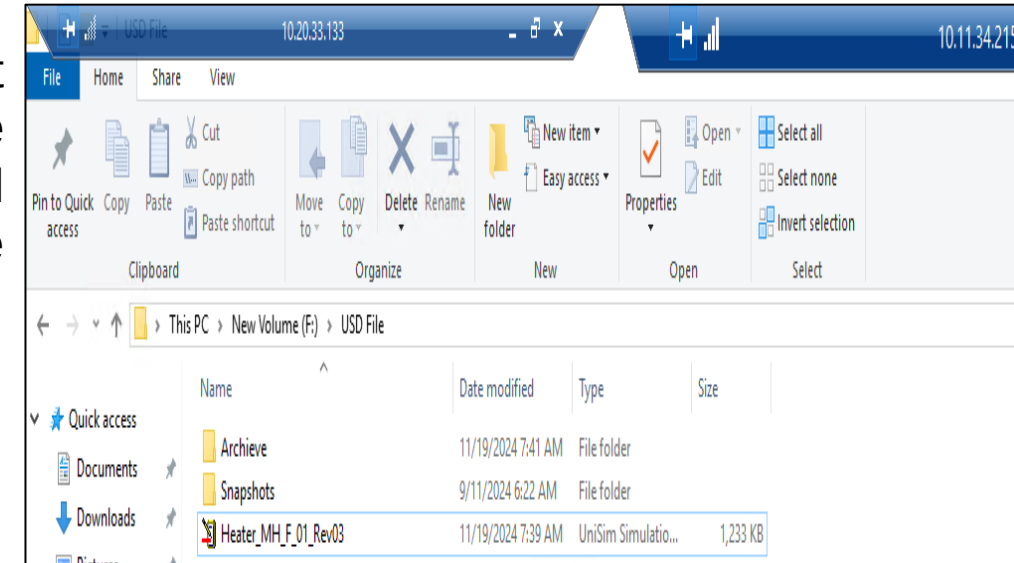
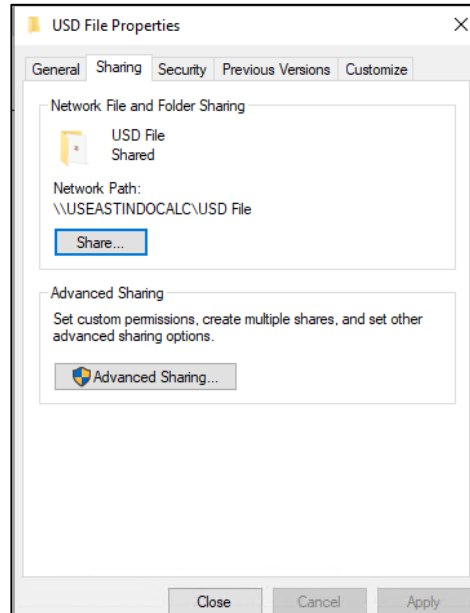
	Object	Variable	Value	Units	Tag	Access Mode
1	Heater calcs	Cell Value (B-2)	1.418e+004	kg/h	BASE_AIR_MASSFLOWRATE	Read/Write
2	Heater calcs	Cell Value (B-3)	1151		BASE_NG_FLOWRATE	Read/Write
3	Heater calcs	Cell Value (B-12)	798.4		BASE_FG_CONVECTIONINLET	Read/Write
4	Heater calcs	Cell Value (B-13)	431.3		BASE_FG_CONVECTIONOUTLE	Read/Write
5	Heater calcs	Cell Value (B-24)	338.8		BASE_PROCESS_OUTLETTEMP	Read/Write
6	Heater calcs	Cell Value (B-29)	312.7		BASE_PROCESS_INTLETTEMP	Read/Write
7	Heater calcs	Cell Value (B-33)	726.5		BASE_PRODUCT_FLOWRATE	Read/Write
8	Heater calcs	Cell Value (B-36)	7.138		BASE_PRODUCT_OUTLETPRES	Read/Write
9	Heater calcs	Cell Value (B-43)	189.8		BASE_AIR_EXITTEMP	Read/Write
10	Heater calcs	Cell Value (B-4)	0.0000		BASE_PROPANE_FLOWRATE	Read/Write
11	Heater calcs	Cell Value (B-5)	178.2		BASE_WASTEGAS_FLOWRATE	Read/Write

UNISIM BUILD INSTALLATION AND CASE CREATION

OUTPUT Data						
	Object	Variable	Value	Units	Tag	Access Mode
1	K-100	Power	1.700	kW	OP_ACT_AIRFANPOWER	Read/Write
2	Heater calcs	Cell Value (F-19)	5.581	%	OP_ACT_EXCESSAIR	Read/Write
3	Heater calcs	Cell Value (F-20)	86.65		OP_ACT_OVERALLEFFICIENCY	Read/Write
4	FH-100	Total Duty	4.371e+007	kJ/h	OP_ACT_TOT_DUTY_PREDICT	Read/Write
5	Heater calcs	Cell Value (B-21)	10.67		ACT_AIRTOFUELRATIO	Read/Write
6	FH-100	Combustion Out Temp	2042	C	OP_ACT_FLAMETEMP	Read/Write
7	Heater calcs	Cell Value (B-32)	12.06	kJ/h	ACT_FUELDUTY	Read/Write
8	E-106	Exchanger Cold Duty	3.468e+007	kJ/h	OP_ACT_RADIATIONDUTY	Read/Write
9	E-102	Exchanger Cold Duty	9.038e+006	kJ/h	OP_ACT_CONVECTIONDUTY	Read/Write
10	Objective func	Cell Value (E-3)	945.8	C	OP_EXP_FLUEGASRADIATIONC	Read/Write
11	Objective func	Cell Value (E-5)	387.0	C	OP_EXP_CONVECTIONOUTTEI	Read/Write
12	Objective func	Cell Value (E-6)	339.5	C	OP_EXP_PROCESSOUTTEMP	Read/Write
13	Heater calcs-2	Cell Value (G-37)	1.386e+007	kJ/h	OP_EXP_CONVECTIONDUTY	Read/Write
14	Heater calcs-2	Cell Value (G-38)	3.158e+007	kJ/h	OP_EXP_RADIATIONDUTY	Read/Write
15	Heater calcs-2	Cell Value (G-39)	4.545e+007	kJ/h	OP_ACT_TOT_DUTY_PREDICT	Read/Write
16	Heater calcs-2	Cell Value (G-40)	104.0		OP_EXP_EFFICIENCY_PREDICT	Read/Write
17	Heater calcs	Cell Value (K-41)	1.006	mol %	ACT_STACKO2	Read/Write
18	Heater calcs	Cell Value (L-41)	1.145	mol %	EXP_EXPECTEDSTACKO2	Read/Write
19	Heater calcs	Cell Value (K-42)	3.227e+006	kJ/h	ACT_APHDUTY	Read/Write
20	Heater calcs	Cell Value (L-42)	3.124e+006	kJ/h	EXP_EXPECTEDAPHDUTY	Read/Write
21	Heater calcs	Cell Value (K-43)	292.9	C	ACT_STACKTEMP	Read/Write
22	Heater calcs	Cell Value (L-43)	251.6	C	EXP_EXPECTEDSTACKTEMP	Read/Write
23	Heater calcs	Cell Value (K-44)	0.3500		ACT_APHBYPASS	Read/Write
24	Heater calcs	Cell Value (L-44)	0.3500		EXP_EXPECTEDAPHBYPASS	Read/Write
25	Heater calcs	Cell Value (L-46)	1.741	kW	OP_EXP_AIRFANPOWER	Read/Write
26	Profit calc	Cell Value (C-21)	4.987e+004		USDPERYEARSAVING	Read/Write
27	Profit calc	Cell Value (C-18)	36.96	kg/h	FUELFLOWREDUCTION	Read/Write
28	Heater calcs-2	Cell Value (E-20)	90.09	%	OP_EXP_EFFICIENCY	Read/Write

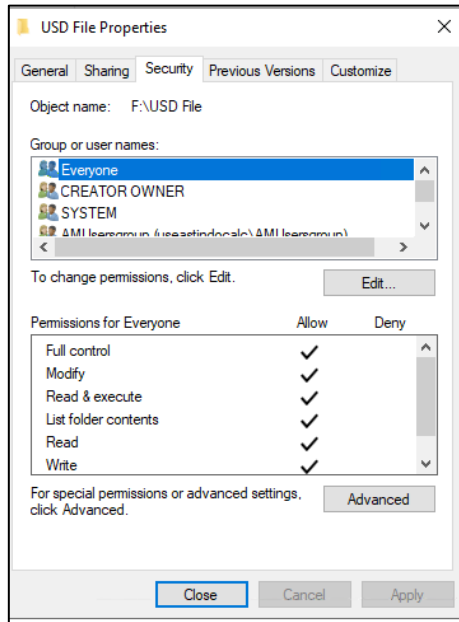
SECURITY AND PERMISSION FOR UNISIM CASE

1. Unisim case should be saved in the calculation server in any folder of drive.
2. Follow below steps to ensure Unisim folder has right permission.
 1. The Asset Performance Management services user must have permission to access the UniSim case file, set at the file level. If permission is not set, or is set at the folder level only, an error will result when attempting to execute the UniSim model.
 2. Unisim file folder should share for services user/Everyone.

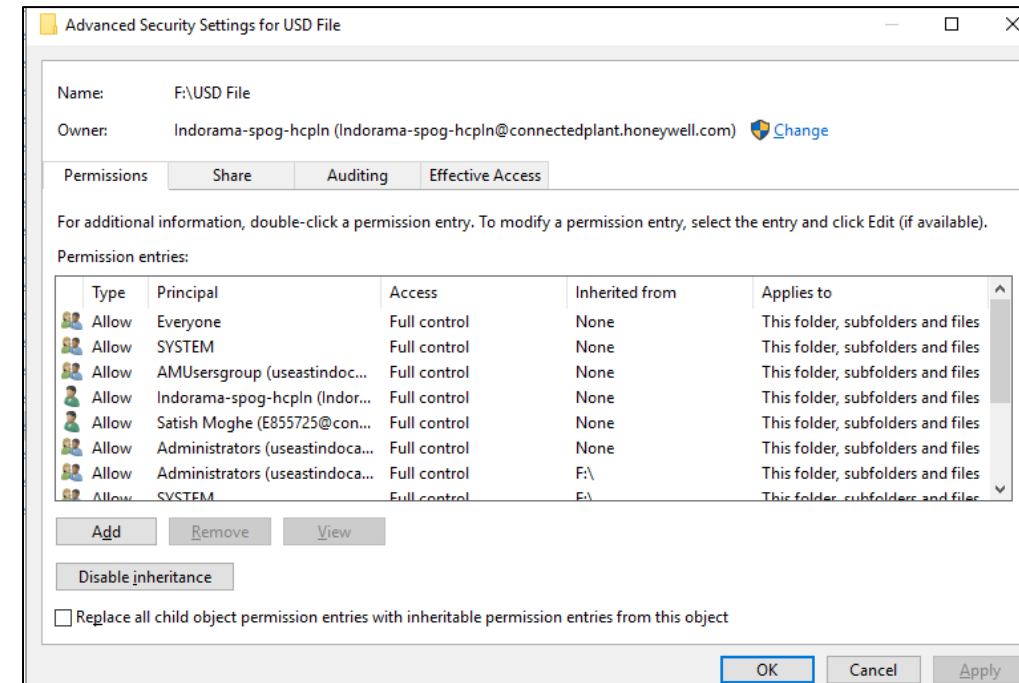
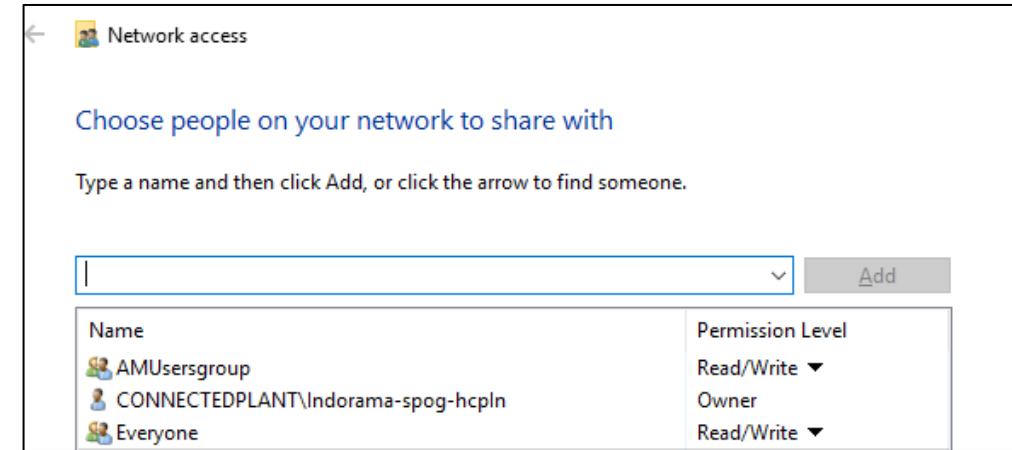


SECURITY AND PERMISSION FOR UNISIM CASE

1. Full control rights should be granted to “AMUsersgroup, Everyone, Service Account” from the Security tab of USD folder.

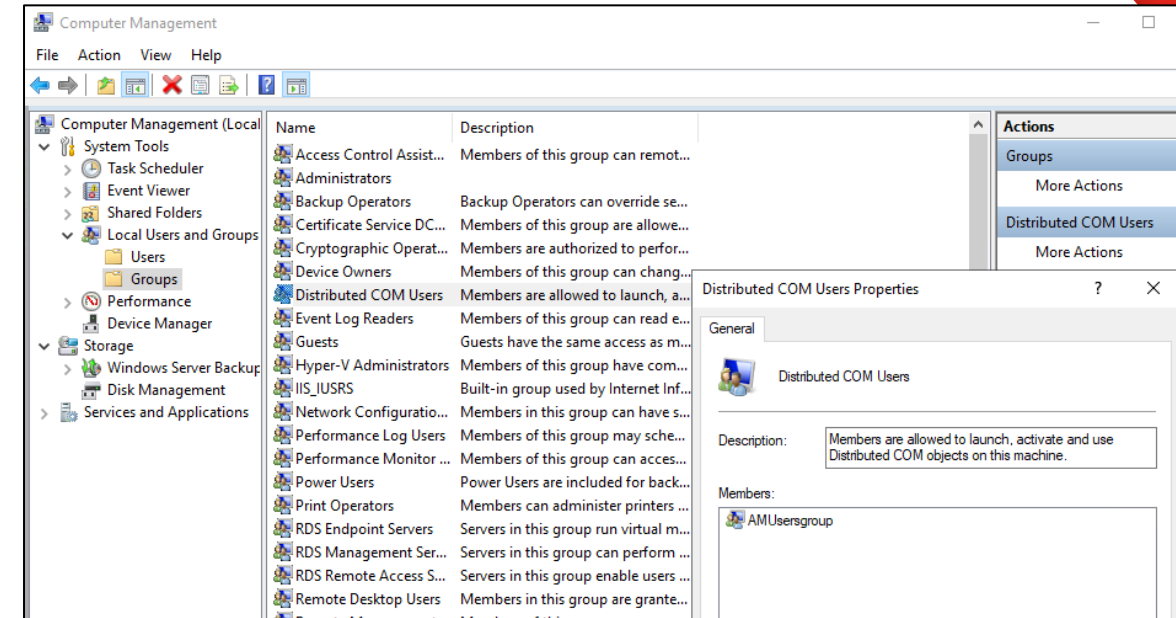
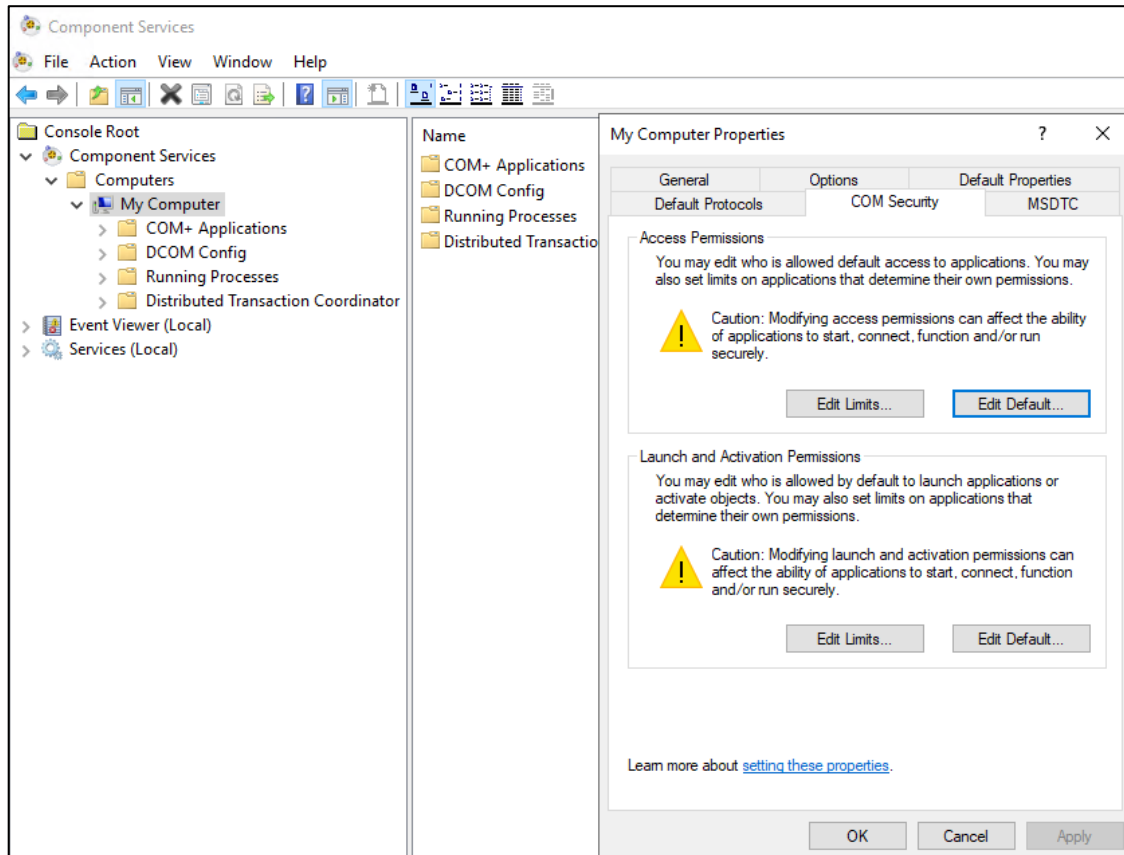


2. From Advance Security Setting, USD file should **Enabled inheritance**.



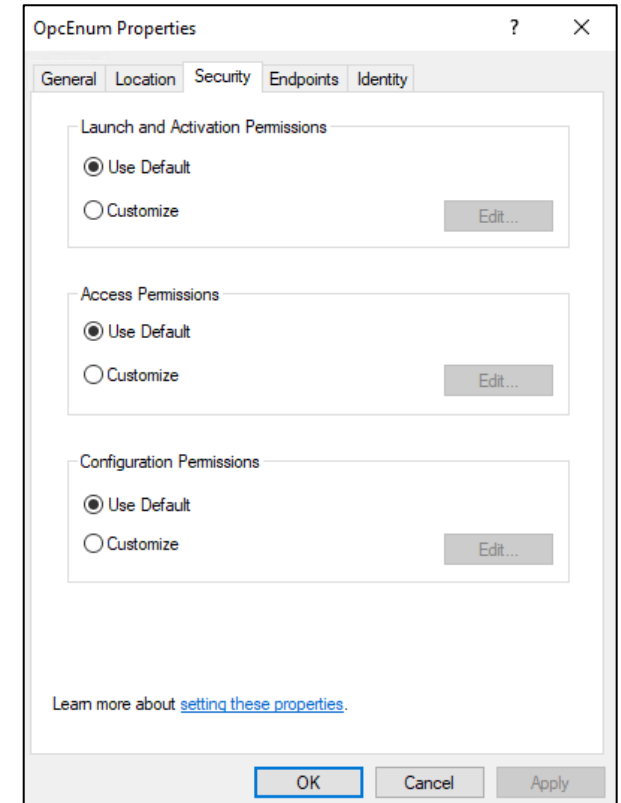
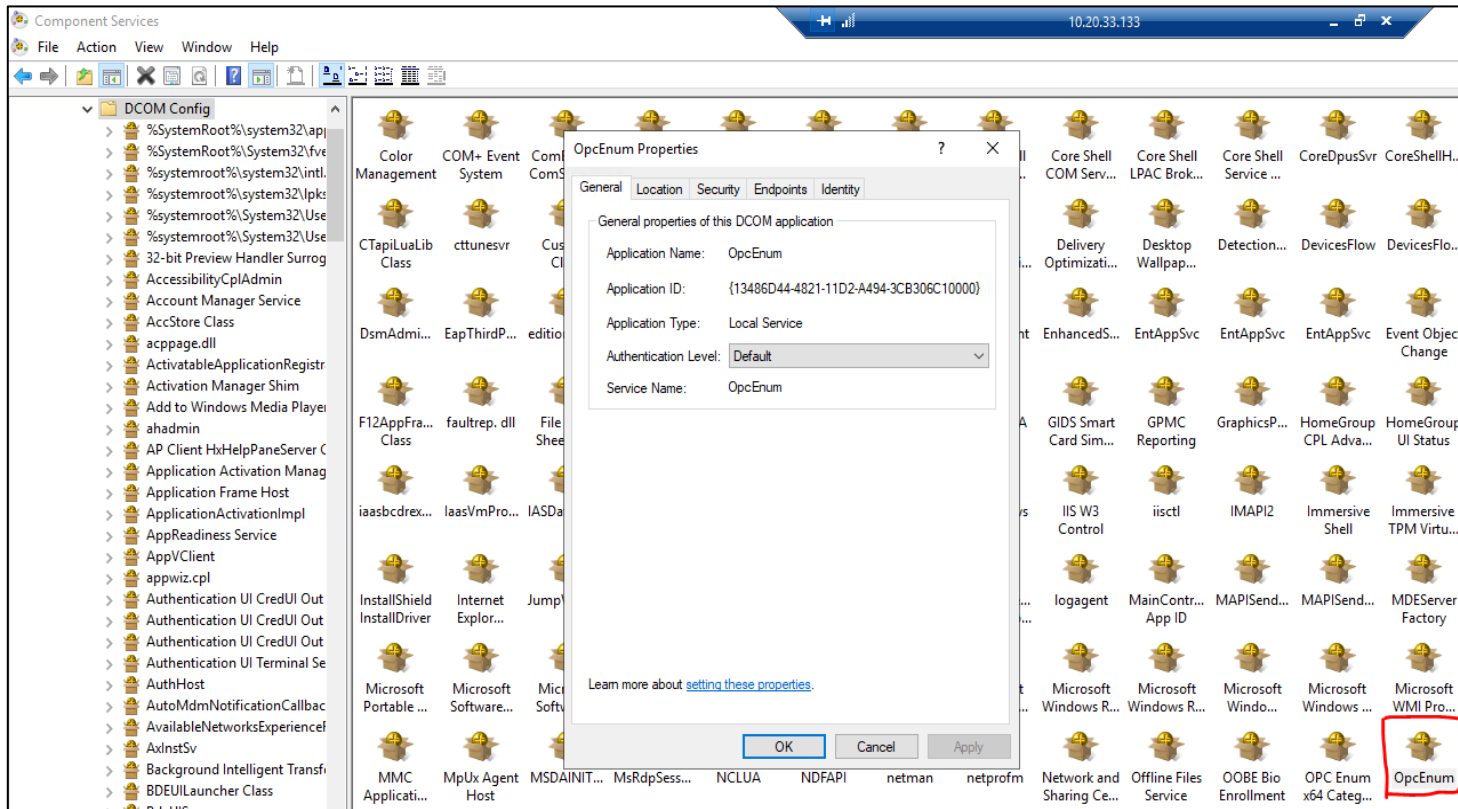
DCOM SETTING ON CALCULATION SERVER:

1. Add AMUsersgroup in Distributed COM Users Group
2. Go to Start > All Apps > Component Services > Computers > My Computer and right click properties > COM Security



DCOM SETTING ON CALCULATION SERVER:

1. Add NETWORK in Access Permissions for Edit Limits and Edit Default
2. Launch and Activation Permissions for Edit Limits and Edit Default. Give all permissions for NETWORK.
3. Go to Start > All Apps > Component Services > Computers > My Computer>DCOM Config> OPCEnum and right click properties > Security
4. Mark “Customize” for Launch and Activation Permission, Access Permission and Configuration Permission.



TO CONFIGURE THE UNISIM DESIGN DATA SOURCE

1. Open Asset Performance Management, and on the Contents tree choose Configuration > Datasources.
2. In the DataSource page, select click Add and select USD: UniSim Design. The USD data source Configuration pane is displayed in the right.
3. In the Data Source Name box, the name of the default data source is displayed. If you want to change the name of the default data source, type a new name for the data source.
4. In the Display Name box, type a name for the data source in the localized language. On the tree view, the name of the data source is displayed in the localized language.

NOTE: If you have a separate calculation server then the input folder and snapshot location is created in the calculation server. Create a shared network folder for the input file and snapshot location and update the folder path. Example: Input File

[CalculationServer]\UnisimModels\USDModel\USC]

\\10.11.34.215\USD File\Heater_MH_F_01_Rev03.usc

Snapshot Location [CalculationServer]\UnisimModels\Snapshots\]

\\10.11.34.215\USD File\Snapshots

The screenshot shows the configuration pane for the USD data source. It includes the following fields and options:

- Name:** USD
- Display Name:** USD EXPECT
- Description:** USD EXPECT
- Type:** USD
- Input File:** \\10.11.34.215\USD File\Heater_MH_F_01_Rev03.usc
- Snap Shot Location:** \\10.11.34.215\USD File\Snapshots
- TimeOut:** 15
- Mode:** Calculation (dropdown menu)
- Data Directions:** Read and Write checkboxes are both checked.
- Buttons:** SAVE and CANCEL buttons are at the bottom.

TO CONFIGURE THE UNISIM DESIGN DATA SOURCE

1. In the Snapshot Location box, type the file path to the folder in which to save the UniSim snapshot, which is the output of the UniSim model execution. Inside this folder location, the snapshots are saved to a Success or Fail folder, depending on the result.
2. In the Timeout box, type the maximum time in minutes the simulation model runs, after which the model is stopped.
3. From the Model list, select one of the following: I Calculation: the calculation of equipment/process performance using the configured UniSim design model. I Optimization: identifying the optimal process condition or equipment parameters using the configured UniSim design model.
4. Click Save to save the USD configuration.
5. Once datasource successfully get saved, Datasource will show as “Not Running”. It needs AMAApplicationServer restart once Input and output tags in Asset Attribute mapping and Unisim has been done else it will not show “Running”.

Name ●
USD
Display Name ●
USD EXPECT
Description
USD EXPECT
Type
USD
Input File ●
\\10.11.34.215\USD File\Heater_MH_F_01_Rev03.usc
Snap Shot Location
\\10.11.34.215\USD File\Snapshots
TimeOut ●
15
Mode
Calculation
Data Directions
Read Write
SAVE
CANCEL

INPUT/OUTPUT VARIABLES

This section describes how to configure Unisim input/output variables with APM input/output variables in Asset Attribute Mapping Sheet.

The following are the high-level tasks to integrate USD with Asset Performance Management

- APM Input Tag will write the data by Writeback Datasource
- For APM Input Tag writeback to UniSim, Write tag name should have prefix **<PDT Input Name><.><Unisim Tag>**. Example **INPUT.BASE_AIR_MOLARFLOWRATE**
- UNISIM Output will read from APM Read Datasource
For Unisim Output tag read to APM, Overridden Tag name should have prefix **<PDT Output Name><.><Unisim Tag>**. Example **OUTPUT.OP_ACT_AIRFANPOWER**
- Map the corresponding input and output tags in Asset Performance Management.

INPUT/OUTPUT VARIABLES

Input from Unisim case

Object	Variable	Value	Units	Tag	Access Mode
Heater calcs	Cell Value (B-2)	19984.00195		BASE_AIR_MOLARFLOWRATE	Read/Write
Heater calcs	Cell Value (B-3)	1418.58667		BASE_FG_FLOWRATE	Read/Write
Heater calcs	Cell Value (B-12)	662.8		BASE_FG_CONVECTIONINLET_TEMP	Read/Write
Heater calcs	Cell Value (B-13)	438.6		BASE_FG_CONVECTIONOUTLET_TEMP	Read/Write
Heater calcs	Cell Value (B-24)	325.35		BASE_PROCESS_OUTLETTEMP	Read/Write
Heater calcs	Cell Value (B-29)	284.377045		BASE_PROCESS_INTLETTEMP	Read/Write
Heater calcs	Cell Value (B-33)	570		BASE_PRODUCT_FLOWRATE	Read/Write
Heater calcs	Cell Value (B-36)	1151		BASE_PRODUCT_OUTLETPRESSURE	Read/Write
Heater calcs	Cell Value (B-43)	250		BASE_AIR_EXITTEMP	Read/Write
Heater calcs-2	Cell Value (B-2)	19984.00195		BASE_AIR_MOLARFLOWRATE	Read/Write
Heater calcs-2	Cell Value (B-3)	1418.58667		BASE_FG_FLOWRATE	Read/Write
Heater calcs-2	Cell Value (B-12)	662.8		BASE_FG_CONVECTIONINLET_TEMP	Read/Write
Heater calcs-2	Cell Value (B-13)	438.6		BASE_FG_CONVECTIONOUTLET_TEMP	Read/Write
Heater calcs-2	Cell Value (B-24)	325.35		BASE_PROCESS_OUTLETTEMP	Read/Write
Heater calcs-2	Cell Value (B-29)	284.377045		BASE_PROCESS_INTLETTEMP	Read/Write
Heater calcs-2	Cell Value (B-33)	570		BASE_PRODUCT_FLOWRATE	Read/Write
Heater calcs-2	Cell Value (B-36)	1034.336792		BASE_PRODUCT_OUTLETPRESSURE	Read/Write
Heater calcs-2	Cell Value (B-43)	271.46		BASE_AIR_EXITTEMP	Read/Write

INPUT/OUTPUT VARIABLES

Input mapping in Asset Attribute Mapping Sheet

Name	Write DataSource	Write Tag Name
Base_Air_MolarFlowRate	USD	INPUT.BASE_AIR_MOLARFLOWRATE
Base_FG_FlowRate	USD	INPUT.BASE_FG_FLOWRATE
Base_FG_ConvectionInlet_Temp	USD	INPUT.BASE_FG_CONVECTIONINLET_TEMP
Base_FG_ConvectionOutlet_Temp	USD	INPUT.BASE_FG_CONVECTIONOUTLET_TEMP
Base_Process_OutletTemp	USD	INPUT.BASE_PROCESS_OUTLETTEMP
Base_Process_IntletTemp	USD	INPUT.BASE_PROCESS_INTLETTEMP
Base_Product_FlowRate	USD	INPUT.BASE_PRODUCT_FLOWRATE
Base_Product_OutletPressure	USD	INPUT.BASE_PRODUCT_OUTLETPRESSURE
Base_Air_ExitTemp	USD	INPUT.BASE_AIR_EXITTEMP
Base_Air_MolarFlowRate	USD	INPUT.BASE_AIR_MOLARFLOWRATE
Base_FG_FlowRate	USD	INPUT.BASE_FG_FLOWRATE
Base_FG_ConvectionInlet_Temp	USD	INPUT.BASE_FG_CONVECTIONINLET_TEMP
Base_FG_ConvectionOutlet_Temp	USD	INPUT.BASE_FG_CONVECTIONOUTLET_TEMP
Base_Process_OutletTemp	USD	INPUT.BASE_PROCESS_OUTLETTEMP
Base_Process_IntletTemp	USD	INPUT.BASE_PROCESS_INTLETTEMP
Base_Product_FlowRate	USD	INPUT.BASE_PRODUCT_FLOWRATE
Base_Product_OutletPressure	USD	INPUT.BASE_PRODUCT_OUTLETPRESSURE
Base_Air_ExitTemp	USD	INPUT.BASE_AIR_EXITTEMP

INPUT/OUTPUT VARIABLES

Output tags in Unisim case

Object	Variable	Value	Unit	Tag	Access Mode
K-100	Power	2.453	kW	OP_ACT_AIRFANPOWER	Read/Write
Heater calcs	Cell Value (F-19)	40.436	%	OP_ACT_EXCESSAIR	Read/Write
Heater calcs	Cell Value (F-20)	73.774		OP_ACT_OVERALLEFFICIENCY	Read/Write
FH-100	Total Duty	39718330	kJ/h	OP_ACT_TOT_DUTY_PREDICT	Read/Write
Heater calcs	Cell Value (B-21)	14.087		ACT_AIRTOFUELRATIO	Read/Write
FH-100	Combustion Out Temperature	1833.907	C	OP_ACT_FLAMETEMP	Read/Write
Heater calcs	Cell Value (B-32)	12.86745	kJ/h	ACT_FUELDUTY	Read/Write
E-106	Exchanger Cold Duty	29702825	kJ/h	OP_ACT_RADIATIONDUTY	Read/Write
E-102	Exchanger Cold Duty	7567076	kJ/h	OP_ACT_CONVECTIONDUTY	Read/Write
Heater calcs-2	Cell Value (F-19)	40.436	%	OP_EXP_EXCESSAIR	Read/Write
FH-100-2	Combustion Out Temperature	1762.27	C	OP_EXP_FLAMETEMP	Read/Write

INPUT/OUTPUT VARIABLES

Output mapping in Asset Attribute Mapping Sheet

Name	Read DataSource	Overridden Tag Name
OP_ACT_AirFanPower	USD	OUTPUT.OP_ACT_AIRFANPOWER
OP_ACT_ExcessAir	USD	OUTPUT.OP_ACT_EXCESSAIR
OP_ACT_OverallEfficiency	USD	OUTPUT.OP_ACT_OVERALLEFFICIENCY
OP_ACT_TOT_DUTY_PREDICT	USD	OUTPUT.OP_ACT_TOT_DUTY_PREDICT
ACT_Airtofuelratio	USD	OUTPUT.ACT_AIRTOFUELRATIO
OP_ACT_FlameTemp	USD	OUTPUT.OP_ACT_FLAMETEMP
ACT_FuelDuty	USD	OUTPUT.ACT_FUELDUTY
OP_ACT_RadiationDuty	USD	OUTPUT.OP_ACT_RADIATIONDUTY
OP_ACT_ConvectionDuty	USD	OUTPUT.OP_ACT_CONVECTIONDUTY
OP_EXP_ExcessAir	USD	OUTPUT.OP_EXP_EXCESSAIR
OP_EXP_FlameTemp	USD	OUTPUT.OP_EXP_FLAMETEMP
OP_EXP_FlueGasRadiationOutTemp	USD	OUTPUT.OP_EXP_FLUEGASRADIATIONOUTTEMP
OP_EXP_CONVECTIONOutTemp	USD	OUTPUT.OP_EXP_CONVECTIONOUTTEMP
OP_EXP_ProcessOutletTemp	USD	OUTPUT.OP_EXP_PROCESSOUTTEMP
OP_EXP_ConvectionDuty	USD	OUTPUT.OP_EXP_CONVECTIONDUTY
OP_EXP_RadiationDuty	USD	OUTPUT.OP_EXP_RADIATIONDUTY

APM-UNISIM CALCULATION SCHEDULER

1. For Unisim execution to trigger, polling interval with desired frequency to be updated manually from DB server for External datasource.
2. Go to DB server
3. Expand the External Data sources table from Asset Sentinel database.
4. Edit the USD datasource by adding Polling interval as below
5. Once done Take AM service restart to effect polling interval and USD datasource for “Running”.

```
<ConnectionXml>  
<Type>USD</Type>  
<InputDirectory>\\10.11.34.215\USD File\Heater_MH_F_01_Rev02.usc</InputDirectory>  
<Mode>0</Mode>  
<OutputDirectory>\\10.11.34.215\USD File\Snapshots</OutputDirectory>  
<Timeout>15</Timeout>  
<PollingInterval>10</PollingInterval>  
<Description>USD  
EXPECT</Description>ConnectionStringProcessBatchIntervalMs=500;MaximumTasks=5;MaximumWaiting  
Tasks=1;MaximumWaitingTimeSeconds=30;MinimumTagsPerBatch=0;MaximumTagsPerBatch=500;Minimu  
mDataPointsPerTag=30;BatchStatisticsSeconds=60;UpdateCountersIntervalMs=1000;WriteFailureRetries=5  
</ConnectionXml>
```


SNAPSHOT FILE – SUCCESS OR FAIL

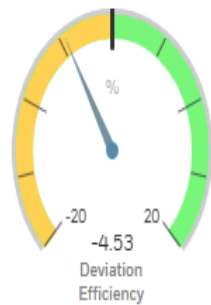
1. Snapshot files to be checked for any Success or Fail along with input file in the Snapshot Folder for unisim model.
2. Input files in the Snapshot folder should be created as per calculation scheduler.
3. Success or Fail file should get created in respective folder as per Input files frequency.
4. For any failed case, Unisim model can be troubleshoot for any errors and to rectify.
5. Input files can check for data written from APM to Unisim correctly.
6. AM logs can check for any missing USD mapping or incorrect configuration.

This PC > New Volume (F:) > USD File > Snapshots > Input			
☐	Name	Date modified	Type
★	Heater_MH_F_01_Rev032024-12-05_03-43-58.usp	12/5/2024 3:46 AM	USP File
★	Heater_MH_F_01_Rev032024-12-05_03-33-58.usp	12/5/2024 3:36 AM	USP File
★	Heater_MH_F_01_Rev032024-12-05_03-20-58.usp	12/5/2024 3:25 AM	USP File
★	Heater_MH_F_01_Rev032024-12-05_03-13-58.usp	12/5/2024 3:15 AM	USP File
★	Heater_MH_F_01_Rev032024-12-05_03-02-58.usp	12/5/2024 3:05 AM	USP File
★	Heater_MH_F_01_Rev032024-12-05_02-53-58.usp	12/5/2024 2:55 AM	USP File
	Heater_MH_F_01_Rev032024-12-05_02-43-58.usp	12/5/2024 2:45 AM	USP File
	Heater_MH_F_01_Rev032024-12-05_02-33-58.usp	12/5/2024 2:35 AM	USP File

New Volume (F:) > USD File > Snapshots > Success			
☐	Name	Date modified	Type
★	Heater_MH_F_01_Rev032024-12-05_03-43-58.usp	12/5/2024 3:46 AM	USP File
★	Heater_MH_F_01_Rev032024-12-05_03-33-58.usp	12/5/2024 3:36 AM	USP File
★	Heater_MH_F_01_Rev032024-12-05_03-20-58.usp	12/5/2024 3:26 AM	USP File
★	Heater_MH_F_01_Rev032024-12-05_03-13-58.usp	12/5/2024 3:16 AM	USP File
★	Heater_MH_F_01_Rev032024-12-05_03-02-58.usp	12/5/2024 3:05 AM	USP File
★	Heater_MH_F_01_Rev032024-12-05_02-53-58.usp	12/5/2024 2:55 AM	USP File
	Heater_MH_F_01_Rev032024-12-05_02-43-58.usp	12/5/2024 2:46 AM	USP File
	Heater_MH_F_01_Rev032024-12-05_02-33-58.usp	12/5/2024 2:35 AM	USP File
	Heater_MH_F_01_Rev032024-12-05_02-22-58.usp	12/5/2024 2:26 AM	USP File
	Heater_MH_F_01_Rev032024-12-05_02-11-58.usp	12/5/2024 2:15 AM	USP File

ENERGY OPPORTUNITY DASHBOARD

1. With Unisim Calculation and Optimization of asset model, energy opportunity \$ Saving calculated.



Paramters	Units	Actual Case	Base Case	Optimize Case
Natural Gas Flow	Nm3/hr	1017.39	1017.39	835.85
Air Flow	Nm3/hr	12939.72	12939.72	11383.36
Fuel Duty	GCal/hr	10.82	10.82	10.07
Air to Fuel Ratio		10.81	10.81	10.25
Efficiency	%	86.88	91.01	89.89

Paramters	Units	Actual Case	Base Case	Optimize Case
Excess Air	%	6.97	6.97	2.00
Convection Outlet Temp	Deg C	417.35	360.67	385.49
Process Outlet Temp	Deg C	332.62	333.49	332.62
Stack O2		1.24	1.38	0.41

Thanks!